# Spotify Songs Analysis

## Data Exploration & Dashboard Development

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# 1. Introduction

## Dataset

| | |
|---|---|
| Dataset title | Most Streamed Spotify Songs 2023 |
| Source | Kaggle |
| Format | CSV (spotify-2023.csv) |
| Size | 953 songs and 24 columns |

**Description:** The dataset contains the most streamed songs on Spotify in 2023. These are real songs by real artists. Each row is one song and includes the song name, artist, release date, number of streams, how many playlists and charts it is in (Spotify, Apple Music, Deezer and Shazam), and music features like danceability, energy, valence (happiness) and bpm (speed).

**Why I selected it:** The data is real and the topic is fun and easy to understand. The data also needs real cleaning (numbers saved as text, missing values, duplicates and a wrong value), and it has both numeric and categorical columns for the charts and the dashboard filters.

## Problem Definition

**What is the dataset about?** Which songs and artists were the most popular on Spotify in 2023, and what makes a song popular.

**Main questions:**

- Q1: Which artist has the most songs in the list?
- Q2: What are the most streamed songs?
- Q3: Are danceable songs streamed more?
- Q4: Are newer songs more common in the list?
- Q5: Are there more Major or Minor songs?

**Expected insights:** I expected famous artists like Taylor Swift to have many songs, new songs to be the most common, and danceable songs to have more streams.

# 2. Data Exploration

The data was loaded with Pandas using `encoding='latin-1'` because some song names have special characters. Then it was checked with `head()`, `shape`, `info()`, `describe()` and `isnull().sum()`.

## Main Columns

| Column | Description | Type when loaded |
|---|---|---|
| track_name | Song name | object |
| artist(s)_name | Artist name(s) | object |
| released_year / month / day | Release date | int64 |
| streams | Total number of streams on Spotify | object (should be a number) |
| in_spotify_playlists | Number of Spotify playlists | int64 |
| in_deezer_playlists | Number of Deezer playlists | object (should be a number) |
| in_shazam_charts | Rank in Shazam charts | object (should be a number) |
| bpm | Speed of the song | int64 |
| key / mode | Musical key and Major/Minor | object |
| danceability_%, energy_%, valence_% ... | Music features from 0 to 100 | int64 |

## Problems Found

| Problem | Column | Count |
|---|---|---|
| Text instead of a number (wrong value) | streams | 1 song |
| Numbers with commas saved as text | in_deezer_playlists | 79 values |
| Numbers with commas saved as text | in_shazam_charts | 7 values |
| Missing values | in_shazam_charts | 50 |
| Missing values | key | 95 |
| Duplicate songs (same name and artist) | track_name, artist(s)_name | 4 |

# 3. Data Cleaning

| Step | What was done |
|---|---|
| Convert streams | Converted streams to numbers. One song had text instead of a number, so it was dropped. |
| Convert playlists / charts | Removed the commas from in_deezer_playlists and in_shazam_charts, then converted them to numbers. |
| Missing in_shazam_charts | Filled with 0, because a missing value means the song is not in the Shazam charts. |
| Missing key | Filled with "Unknown". We cannot know the key, and dropping 95 songs (10%) would lose a lot of data. |
| Remove duplicates | Removed 4 songs that appeared twice with the same name and artist. |
| Rename columns | Renamed columns to simple names (artist(s)_name to artist_name, danceability_% to danceability ...). |
| Save | Saved the result as clean_data.csv. |

## Result

| | Before cleaning | After cleaning |
|---|---|---|
| Rows (songs) | 953 | 948 |
| Missing values | 145 | 0 |
| Numeric columns saved as text | 3 | 0 |
| Duplicate songs | 4 | 0 |

Only 5 songs were removed (1 wrong value and 4 duplicates), so almost all the data was kept.

# 4. Exploratory Data Analysis (EDA)

## Univariate Analysis

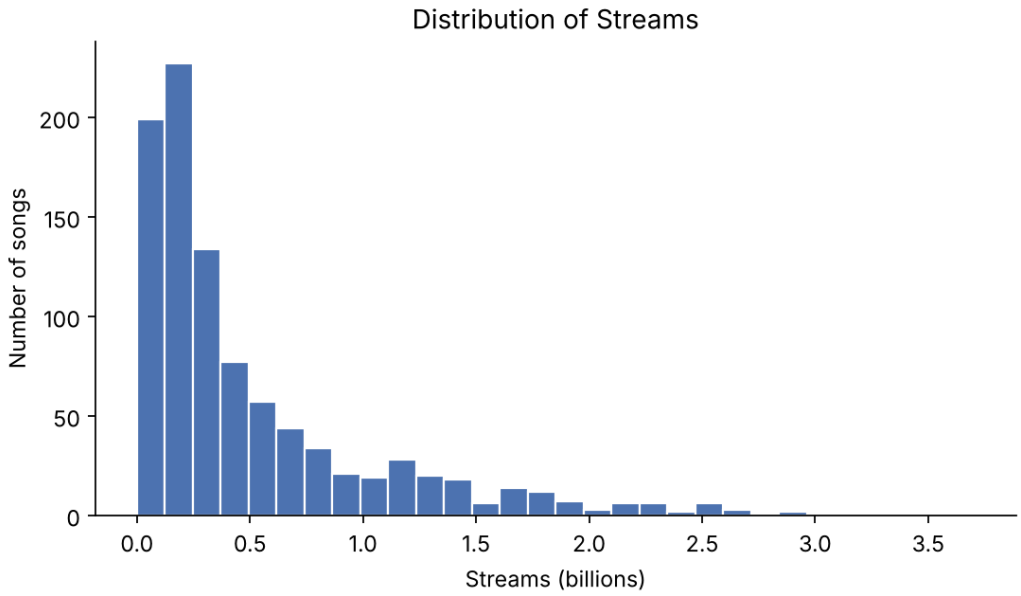



*Figure 1: Histogram of streams. Most songs have less than 1 billion streams. The data is right-skewed: the median is about 288 million and the mean is about 514 million.*

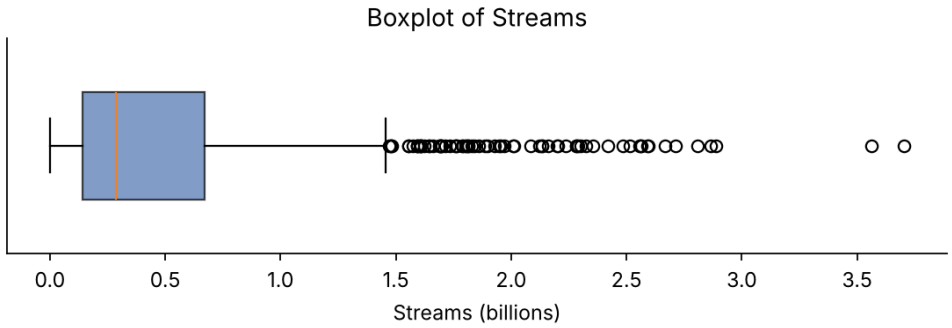



*Figure 2: Boxplot of streams. There are 74 outliers above about 1.47 billion streams. They are real hit songs like Blinding Lights, so they were kept.*

### Q1: Which artist has the most songs in the list?

**Answer:** Taylor Swift has the most songs (34), then The Weeknd (21), and Bad Bunny and SZA (19 each).

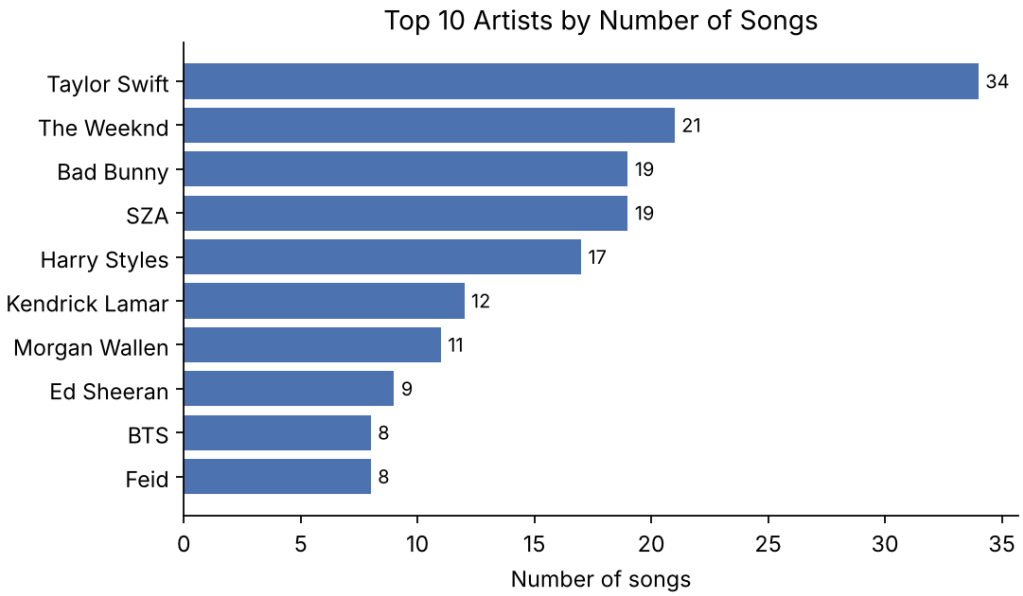



*Figure 3: Top 10 artists by number of songs.*

### Q4: Are newer songs more common in the list?

**Answer:** Yes. About 77% of the songs were released in 2020 or later, and 2022 has the most songs (399).

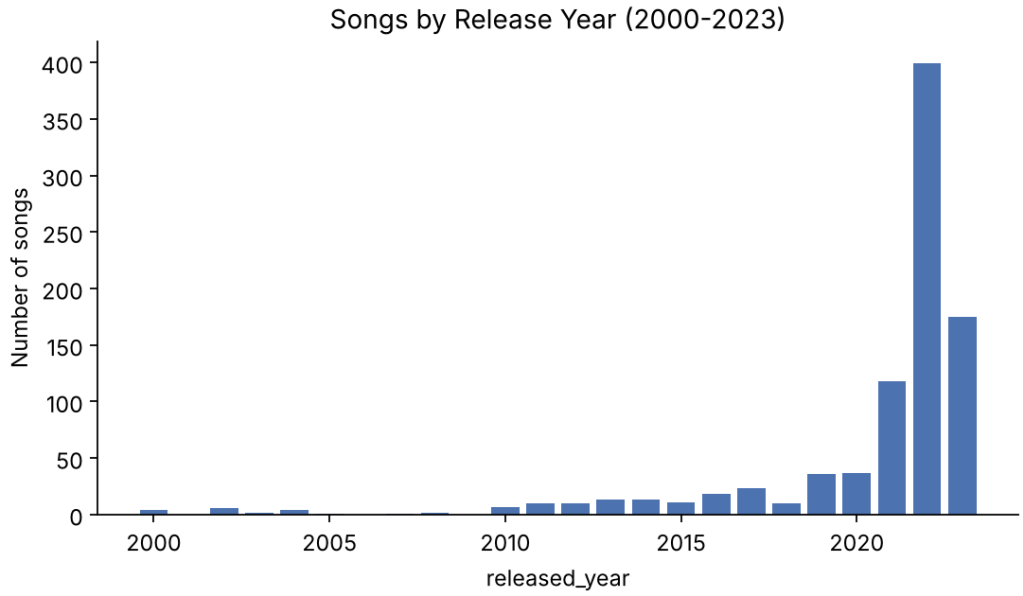



*Figure 4: Number of songs by release year (2000-2023).*

### Q5: Are there more Major or Minor songs?

**Answer:** Major songs are more common (546) than Minor songs (402). Major songs also have a little more streams on average (535 million vs 485 million).

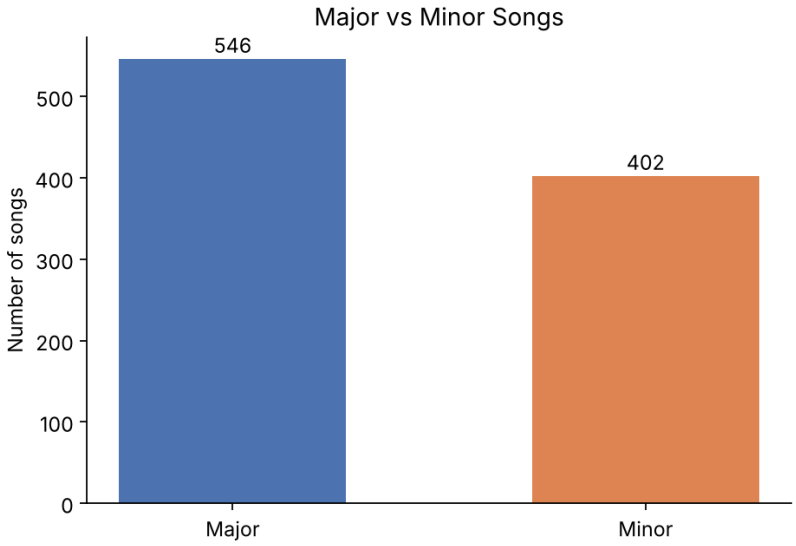



*Figure 5: Number of Major and Minor songs.*

## Bivariate & Multivariate Analysis

### Q2: What are the most streamed songs?

**Answer:** Blinding Lights by The Weeknd is the most streamed song (3.70 billion), then Shape of You by Ed Sheeran (3.56 billion). All of the top 10 songs were released between 2016 and 2021, so older songs had more time to collect streams.

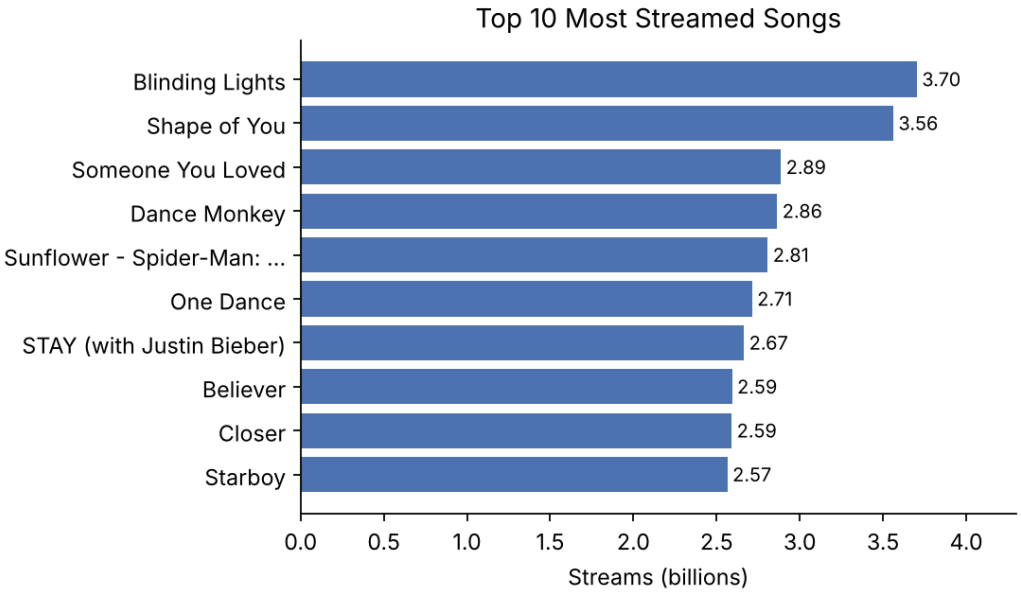



*Figure 6: Top 10 most streamed songs.*

## Q3: Are danceable songs streamed more?

**Answer:** No. There is no clear relation between danceability and streams (correlation -0.11).

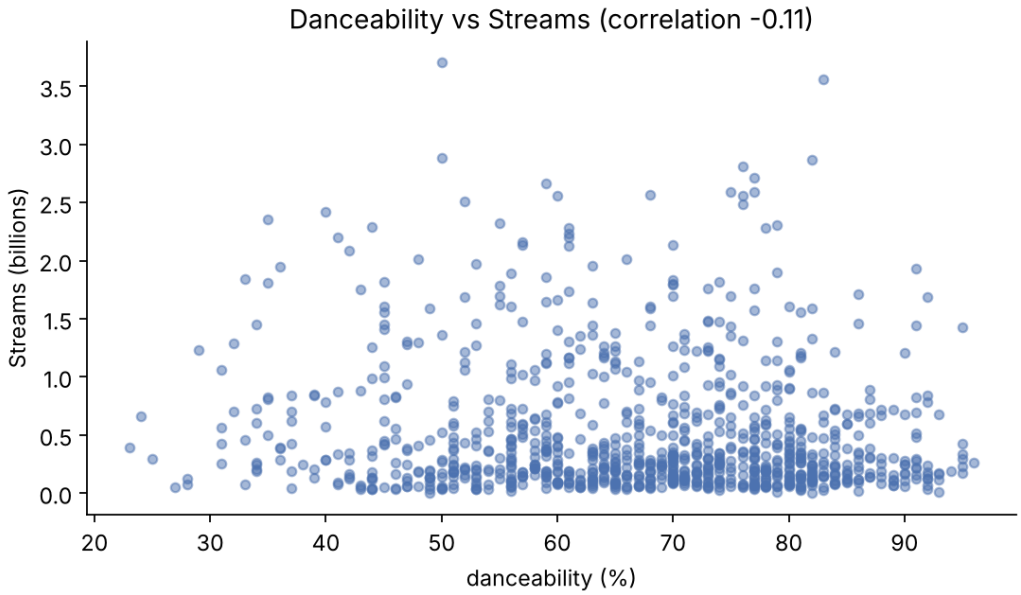



*Figure 7: Scatter plot of danceability vs streams.*

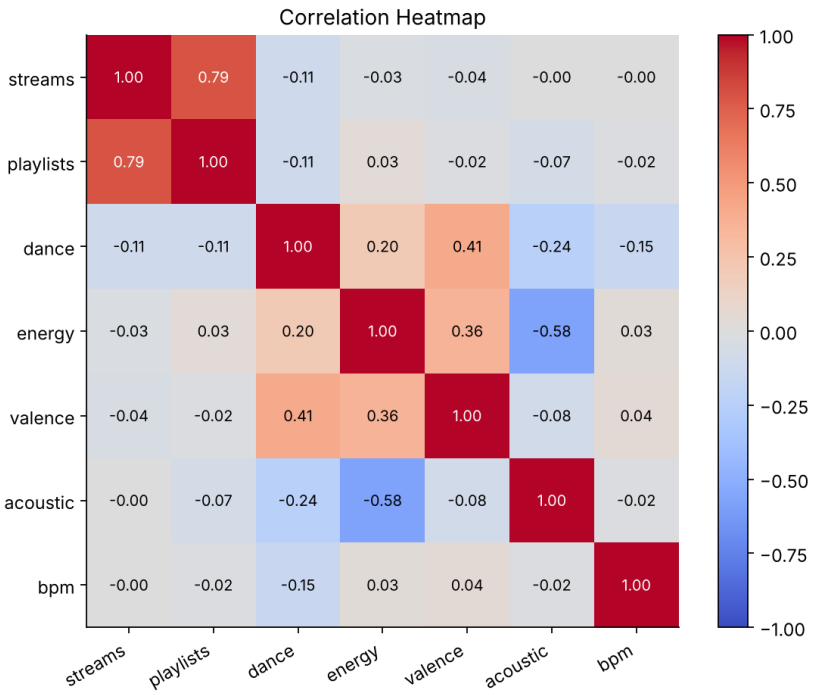



*Figure 8: Correlation heatmap. Streams is strongly related to the number of Spotify playlists (0.79). The music features have almost no relation with streams. Energy and acousticness are negatively related (-0.58).*

# 5. Dashboard Overview

An interactive dashboard was built with Streamlit (app.py). It reads clean_data.csv.

**Dashboard link:** https://spotify-dashboard-vxewipumrcs9rua33n6zcr.streamlit.app

## Sidebar

- Dataset description.
- **Dropdown** to choose the mode (All, Major or Minor).
- **Slider** to choose the release year range.
- **Multiselect** to choose artists (top 20 artists).

## Main Page

- Data preview of the filtered songs.
- Summary statistics: number of songs, total streams, average danceability and describe().
- Interactive charts: top artists, top songs, songs by release year, danceability vs streams.
- Insights section.

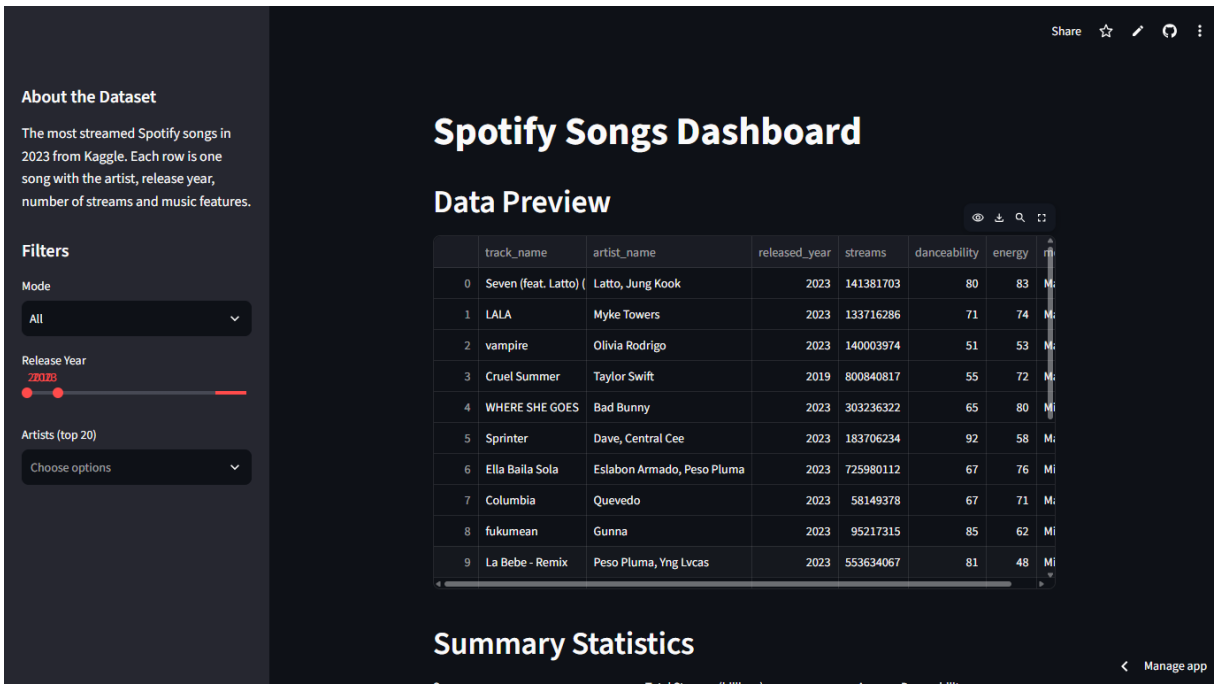



*Figure 9: The Streamlit dashboard with the sidebar filters and the data preview.*

# 6. Insights & Conclusion

## Main Insights

- Taylor Swift has the most songs in the list (34).
- Blinding Lights by The Weeknd is the most streamed song (3.70 billion streams).
- Most songs in the list are new: about 77% were released in 2020 or later.
- Major songs are more common than Minor songs.
- Music features (danceability, energy, valence, bpm) do not affect the number of streams.
- Songs in more Spotify playlists have more streams (correlation 0.79).

## What I Learned

- How to find numbers saved as text and convert them.
- How to decide between filling and dropping missing values.
- How to choose the right chart for each question.
- How to build a Streamlit dashboard with filters.

## Limitations of the Dataset

- The data is only for 2023, so it does not show changes over the years.
- Older songs had more time to collect streams, so comparing streams between old and new songs is not fair.
- The artist column mixes solo songs and songs with more than one artist.
- 95 songs have an unknown key.

## Recommendations for Future Analysis

- Use data from more years to compare trends.
- Compare streams per day since release instead of total streams.
- Split songs with more than one artist to count each artist.
- Add the music genre to compare genres.

# 7. References


- Dataset: Most Streamed Spotify Songs 2023, Kaggle. https://www.kaggle.com/datasets/nelgiriyewithana/top-spotify-songs-2023
- GitHub repository: https://github.com/Razan-2/spotify-dashboard
- Streamlit dashboard: https://spotify-dashboard-vxewipumrcs9rua33n6zcr.streamlit.app
- Pandas documentation: https://pandas.pydata.org/docs/
- Seaborn documentation: https://seaborn.pydata.org/
- Streamlit documentation: https://docs.streamlit.io/